

Heinrich Weber
Textbook of Algebra. Volume I.
Fourth Section. Symmetric Functions.
§ 46 end–§ 54

§ 46. Discriminants

Here, according to § 42, one is to set

$$\begin{aligned} s_0 &= 3, & a_0 s_1 &= -a_1, \\ a_0^2 s_2 &= a_1^2 - 2a_0 a_2, \\ a_0^3 s_3 &= -a_1^3 + 3a_0 a_1 a_2 - 3a_0^2 a_3, \\ a_0^4 s_4 &= a_1^4 - 4a_0 a_1^2 a_2 + 4a_0^2 a_1 a_3 + 2a_0^2 a_2^2, \end{aligned}$$

from which one obtains

$$D = a_1^2 a_2^2 + 18a_0 a_1 a_2 a_3 - 4a_0 a_2^3 - 4a_1^3 a_3 - 27a_0^2 a_3^2. \quad (10)$$

§ 47. Discriminants of forms of the third and fourth orders

For functions of the third and fourth orders, the solutions of the equations of these two degrees (§ 35, 37) give us a simple means for computing the discriminants.

First, for the equation of the third order, the three roots of

$$x^3 + ax + b = 0 \quad (1)$$

were represented in the form

$$\begin{aligned} \alpha &= u + v, \\ \beta &= \varepsilon u + \varepsilon^2 v, \\ \gamma &= \varepsilon^2 u + \varepsilon v, \end{aligned} \quad (2)$$

where

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \quad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{2} \quad (3)$$

are the complex third roots of unity, and u, v are determined by § 35, (8). From (3) one obtains

$$\varepsilon - \varepsilon^2 = \sqrt{-3}, \quad 1 - \varepsilon = \varepsilon^2 \sqrt{-3}, \quad 1 - \varepsilon^2 = -\varepsilon \sqrt{-3}, \quad (4)$$

and from (2)

$$\begin{aligned} \alpha - \beta &= \sqrt{-3}(\varepsilon^2 u - \varepsilon v), \\ \alpha - \gamma &= -\sqrt{-3}(\varepsilon u - \varepsilon^2 v), \\ \beta - \gamma &= \sqrt{-3}(u - v). \end{aligned}$$

Hence, by multiplication,

$$(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma) = 3\sqrt{-3}(u^3 - v^3). \quad (5)$$

Squaring, one obtains for the discriminant of the cubic equation (1), according to § 35, (6),

$$D = -(27b^2 + 4a^3). \quad (6)$$

From this one obtains the discriminant of the general function

$$a_0 x^3 + a_1 x^2 + a_2 x + a_3 \quad (7)$$

by putting for a and b the expressions of § 34, (4), replacing a_1, a_2, a_3 there by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0},$$

and multiplying by a_0^4 [§ 46, (3)]. This gives

$$a = \frac{3a_0a_2 - a_1^2}{3a_0^2}, \quad b = \frac{2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3}{27a_0^3},$$

and consequently

$$D = -\frac{1}{27a_0^2} \left\{ (2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3)^2 + 4(3a_0a_2 - a_1^2)^3 \right\}. \quad (8)$$

If one carries out the square and the cube, one returns to formula (10) of the preceding paragraph; it is noteworthy, however, that formula (8) apparently contains the denominator $27a_0^2$, which cancels when the expansion is carried out.

In exactly the same way we may proceed for biquadratic equations. If $\alpha, \beta, \gamma, \delta$ are the roots of the biquadratic equation

$$x^4 + ax^2 + bx + c = 0, \quad (9)$$

then, by § 37, (5), we obtain

$$\begin{aligned} \alpha - \beta &= v + w, & \gamma - \delta &= v - w, \\ \alpha - \gamma &= w + u, & \delta - \beta &= w - u, \\ \alpha - \delta &= u + v, & \beta - \gamma &= -u + v. \end{aligned} \quad (10)$$

Accordingly the discriminant of equation (9) is

$$D = (v^2 - w^2)^2(w^2 - u^2)^2(u^2 - v^2)^2,$$

that is, D is equal to the discriminant of the cubic resolvent § 37, (4),

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \quad (11)$$

whose roots are u^2, v^2, w^2 .

If, however, this discriminant is formed by formula (10), § 46, with

$$a_0 = 1, \quad a_1 = 2a, \quad a_2 = a^2 - 4c, \quad a_3 = -b^2,$$

one obtains

$$D = -4a^3b^2 + 144acb^2 + 16a^4c - 128a^2c^2 + 256c^3 - 27b^4. \quad (12)$$

If, on the other hand, formula (8) is used to form the discriminant of the cubic equation (11), one obtains

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2. \quad (13)$$

From this one derives the discriminant of the general fourth-degree function

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 \quad (14)$$

by expressing a, b, c , according to § 34, (4), by

$$-\frac{3a_1^2}{8} + a_2, \quad \frac{a_1^3}{8} - \frac{a_1a_2}{2} + a_3, \quad -\frac{3a_1^4}{256} + \frac{a_1^2a_2}{16} - \frac{a_1a_3}{4} + a_4,$$

then replacing a_1, a_2, a_3, a_4 by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0}, \quad \frac{a_4}{a_0},$$

and multiplying by a_0^6 . After carrying out this simple calculation, set, for abbreviation,

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \quad (15)$$

$$B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3. \quad (16)$$

Then (13) gives

$$27D = 4A^3 - B^2. \quad (17)$$

The quantities A and B , which will occupy us further in the next section, are called the first and second invariants of the biquadratic function f . Thus, by (17), one may express the discriminant rationally through these quantities, though not in such a way that the coefficients become integral. If the calculation prescribed in (17) is carried out in order to represent D itself through the coefficients a_0, a_1, a_2, a_3, a_4 , then the factor 27 must still cancel. We shall not write down here the long expression, whose calculation offers no difficulty.

§ 48. Resultants

Another application of the theory of symmetric functions, which also contains the formation of discriminants as a special case, is the formation of resultants.

Let

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n, \quad (1)$$

$$\varphi(x) = b_0x^m + b_1x^{m-1} + b_2x^{m-2} + \cdots + b_m \quad (2)$$

be two integral rational functions of degrees n and m , and let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the roots of the first, so that one may write

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n). \quad (3)$$

Now the product

$$\varphi(\alpha_1)\varphi(\alpha_2) \cdots \varphi(\alpha_n)$$

is a symmetric function of the roots of $f(x)$ and hence can be expressed integrally and rationally through

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \dots, \frac{a_n}{a_0}.$$

The order of this symmetric function is $(m, m \dots m)$ (§ 45), and therefore

$$R = a_0^m \varphi(\alpha_1)\varphi(\alpha_2) \cdots \varphi(\alpha_n) \quad (4)$$

is an integral rational and homogeneous function of order m in $a_0, a_1, \dots, a_n$. It is also, as its expression shows, an integral homogeneous function of order n in $b_0, b_1, \dots, b_m$. It is called the resultant of the two functions $f(x)$ and $\varphi(x)$. This designation is justified by a second representation.

Let the roots of $\varphi(x)$ be denoted by $\beta_1, \beta_2, \dots, \beta_m$, so that

$$\varphi(x) = b_0(x - \beta_1)(x - \beta_2) \cdots (x - \beta_m). \quad (5)$$

Then, by (4),

$$R = a_0^m b_0^n \prod_{i,k} (\alpha_i - \beta_k), \quad (6)$$

where the product sign $\prod$ refers to all indices $i = 1, 2, \dots, n$, $k = 1, 2, \dots, m$.

This expression now shows that R , apart from the factor $(-1)^{mn}$, depends on $\varphi(x)$ in the same way as on $f(x)$; hence one can also set

$$(-1)^{mn} R = b_0^n f(\beta_1)f(\beta_2) \cdots f(\beta_m). \quad (7)$$

Accordingly, discriminants are a special case of resultants, obtained by taking $\varphi(x)$ to be the derivative $f'(x)$.

The vanishing of the resultant of $f(x)$ and $\varphi(x)$ is the necessary and sufficient condition that the two equations

$$f(x) = 0 \quad \text{and} \quad \varphi(x) = 0$$

have a common root.

The equation obtained by setting the resultant equal to zero is also called the final equation from $f(x) = 0$ and $\varphi(x) = 0$, and the formation of the final equation is the elimination of x .

For the computation of resultants one may use the algorithm of the greatest common divisor (§ 6). Namely, if $m \leq n$, then one can determine an integral function Q and likewise a function $\psi(x)$, of degree lower than m , such that

$$f(x) = Q\varphi(x) + \psi(x).$$

Thus, putting x equal to one of the roots β_i ,

$$f(\beta_i) = \psi(\beta_i),$$

and hence

$$b_0^n \prod_i f(\beta_i) = b_0^n \prod_i \psi(\beta_i). \quad (8)$$

If m' is the degree of $\psi(x)$, then

$$b_0^{m'} \prod_i \psi(\beta_i) = R' \quad (9)$$

is the resultant of $\varphi(x)$ and $\psi(x)$, and therefore

$$R = b_0^{n-m'} R'. \quad (10)$$

The formation of the resultant R is thereby reduced to the formation of R' . One can now divide $\varphi(x)$ again by $\psi(x)$, and so reduce the formation of R' to the formation of a resultant of functions of ever lower degree, until one finally reaches a constant, that is, a function of the coefficients a, b , which must be identical with the resultant sought.

The computation of resultants is usually quite lengthy. We shall record only the simplest example, the resultant of two quadratic functions:

$$f(x) = a_0x^2 + a_1x + a_2, \quad \varphi(x) = b_0x^2 + b_1x + b_2.$$

Here one may directly compute the product

$$R = a_0^2(b_0\alpha_1^2 + b_1\alpha_1 + b_2)(b_0\alpha_2^2 + b_1\alpha_2 + b_2),$$

and obtains, after setting

$$\begin{aligned} -a_0(\alpha_1 + \alpha_2) &= a_1, & a_0\alpha_1\alpha_2 &= a_2, \\ R &= a_0^2b_2^2 + a_2^2b_0^2 - b_0b_1a_1a_2 - a_0a_1b_1b_2 + b_0b_2a_1^2 + a_0a_2b_1^2 - 2a_0b_0a_2b_2, \end{aligned} \quad (11)$$

or, equivalently,

$$R = (a_0b_2 - a_2b_0)^2 - (a_0b_1 - a_1b_0)(a_1b_2 - a_2b_1). \quad (12)$$

The individual terms of the resultant of the two functions $f(x)$ and $\varphi(x)$ of degrees n and m have, apart from a numerical integral factor, the form

$$a_0^{\nu_0} a_1^{\nu_1} \cdots a_n^{\nu_n} b_0^{\mu_0} b_1^{\mu_1} \cdots b_m^{\mu_m}, \quad (13)$$

where, as follows from the degrees determined above,

$$\nu_0 + \nu_1 + \cdots + \nu_n = m, \quad \mu_0 + \mu_1 + \cdots + \mu_m = n. \quad (14)$$

But we can state still another relation among these exponents. Since, by (6), R is a homogeneous function of degree nm in the $n + m$ variables α, β , and since further a_0 is of degree zero, a_1 of degree one, a_2 of degree two, and so on in the α , and in general a_k and b_k are of degree k in these variables, it follows that

$$\nu_1 + 2\nu_2 + 3\nu_3 + \cdots + n\nu_n + \mu_1 + 2\mu_2 + 3\mu_3 + \cdots + m\mu_m = nm. \quad (15)$$

This is a relation from which an important conclusion follows, one that we shall discuss somewhat more fully in the following paragraph.

§ 49. Elimination. Bezout's theorem

Suppose that in the two functions

$$\begin{aligned} f(x) &= a_0x^n + a_1x^{n-1} + \cdots + a_n, \\ \varphi(x) &= b_0x^m + b_1x^{m-1} + \cdots + b_m \end{aligned} \quad (1)$$

the coefficients a, b are themselves again integral rational functions of a quantity y . Then the resultant will also become an integral rational function of y , which we shall denote by $F(y)$. The formation of this equation is the elimination of x from the two equations (1).

The equation

$$F(y) = 0 \quad (2)$$

has as roots all those values of y for which the two equations

$$f(x) = 0, \quad \varphi(x) = 0 \quad (3)$$

have a common root. If condition (2) is satisfied, one finds the values of x satisfying the two equations (3) by seeking the greatest common divisor of $f(x)$ and $\varphi(x)$. This common divisor is of the first degree and hence determines x rationally when there is only one such common root; otherwise it is of correspondingly higher degree, and the determination of x still requires the solution of an equation of higher degree.

If only one common root x belongs to each root of the resultant, then the number of value-pairs x, y satisfying equations (3) is equal to the degree of the resultant with respect to y . Thus, if the coefficients a are of degree ν , and the coefficients b of degree μ , with respect to y , then by the degree determination of the preceding paragraph the degree of the resultant is

$$n\nu + m\mu; \quad (5)$$

this therefore is the number of common root-pairs. This number can, however, still be modified by the possibility that the coefficients are so constituted that the highest power of y disappears from the final equation. One would then be able to save the agreement of the number with (5) only by arbitrarily adding infinitely distant roots. Multiple roots also raise difficulties. One partly avoids these inconveniences by taking as the basis the homogeneous form of the equations, a form in which the elimination problem arises especially naturally in geometry.

Namely, if the function $f(x)$ has arisen from a homogeneous function of order n in the three variables x, y, z (a ternary form) by arranging it according to powers of x , then the coefficients $a_0, a_1, a_2, \dots, a_n$ are homogeneous forms in the two variables y, z , of the degree indicated by the index; thus a_0 is a constant, a_1 a linear form, a_2 a quadratic form, and so forth. The same applies to the coefficients $b_0, b_1, \dots, b_m$ of the function $\varphi(x)$.

If x, y, z denote triangular coordinates in the plane, then $f(x) = 0, \varphi(x) = 0$ are the equations of curves of orders n and m , and the common roots of these two equations, that is the ratios $x : y : z$, denote the points of intersection of the two curves. The final equation obtained by eliminating x is a homogeneous equation whose degree follows from (15), § 48 as mn .

A diminution of the degree cannot occur here; the final equation is always a homogeneous equation of degree nm in the two unknowns y, z . The only exception to which one must attend is the case in which the resultant vanishes identically (for all y, z), so that infinitely many common roots are present. Geometrically, this case means that the two curves have a common curve-part and not merely individual points in common.

If the equations $f = 0$, $\varphi = 0$ have only a finite number of common roots, then in the expressions

$$y_1 = y + \beta x, \quad z_1 = z + \gamma x$$

we can choose the constants β, γ so that, if to one value of the ratio $y : z$ there correspond several different values of x satisfying equations (1), then the corresponding values of $y_1 : z_1$ come out different from one another.

We may then regard f and φ also as homogeneous functions of degrees m and n in the variables x, y_1, z_1 , and the resultant then becomes a homogeneous function of degree mn , $R(y_1, z_1)$, in y_1 and z_1 . If in it we put

$$y_1 = b' + b\lambda, \quad z_1 = c' + c\lambda,$$

then we obtain an equation of degree mn in λ , and, provided that $R(y_1, z_1)$ does not vanish identically, we may choose the constants b, b', c, c' so that the coefficient of the highest power of λ , namely $R(b, c)$, is non-zero. Then to each root of the resultant there corresponds only one system of values of the ratio $x : y : z$, and the number of common roots of f and φ is at most equal to mn . To be able to state the theorem that the number of common roots is always equal to mn , one must still make a convention according to which certain value-pairs are to be counted more than once.

The theorem which, in geometrical garb, says that two curves of orders m and n meet in mn points, is called Bezout's theorem.

§ 50. Elimination from several equations

The principles of elimination that we have applied in the preceding discussion to two equations may be extended without difficulty to several equations with a corresponding number of unknowns.

First take the equations

$$f(x) = 0, \quad \varphi(x) = 0, \quad \psi(x) = 0, \quad (1)$$

of degrees m, n, p , whose coefficients a_i, b_i, c_i are for the present to be regarded as independent variables.

We apply the algorithm of the greatest common divisor to two of them, say to $f(x)$ and $\varphi(x)$, and obtain a last remainder which is a rational function of the a and b and, after clearing denominators, agrees with the resultant R of $f(x)$ and $\varphi(x)$, and a next-to-last remainder which is a linear function of x . The condition for the existence of a common root of $f = 0$ and $\varphi = 0$ is the vanishing of the resultant; under this condition the next-to-last remainder gives a rational expression for the common root in terms of the coefficients.

If one substitutes this expression in $\psi(x)$, then, after all denominators have been removed, one obtains an integral rational function of the coefficients a, b, c , which we shall denote by S ; its vanishing, together with $R = 0$, contains the necessary and sufficient condition that the three equations (1) have a common root.

Now replace the coefficients a, b, c of the functions (1) by integral rational functions of a second unknown y , whose coefficients will be denoted by α, β, γ and for the present may be regarded as independent variables. Then R and S become integral rational functions of α, β, γ and y , and if, as equations in y , we consider

$$R = 0, \quad S = 0, \quad (2)$$

then, according to the rules of the preceding paragraph, we may form their resultant T . If it does not vanish identically, this will be an integral rational function of the coefficients α, β, γ , and the equation

$$T = 0 \quad (3)$$

is the necessary and sufficient condition that the system of equations (1) can be satisfied by one or more value-pairs x, y . T is called the resultant of the system (1). Thus one may form from the three equations (1), by eliminating the unknowns x and y , a final equation for z .

If the α, β, γ are now replaced again by integral rational functions of a third unknown z , then T passes into an equation in z . The functions f, φ, ψ become functions of the three unknowns x, y, z . The degree of equation (3) gives the number of systems of values x, y, z for which the three equations (1) are simultaneously satisfied.

If, however, the coefficients are so constituted that T vanishes identically for all values of z , then there are infinitely many systems of values x, y, z simultaneously satisfying equations (1). If one regards x, y, z as Cartesian coordinates, then equations (1) each represent a surface. The degree of the final equation $T = 0$ with respect to z gives the number of intersection points of these three surfaces. If, however, T is identically zero, the three surfaces have either a common surface part or a common curve.

The degree of the final equation can be determined in the following way, which requires no lengthy considerations about the degrees of the functions R, S, T . To make ourselves independent of exceptional cases, we introduce homogeneous variables by adjoining a fourth variable u , and thus consider three homogeneous quaternary equations

$$F(x, y, z, u) = 0, \quad \Phi(x, y, z, u) = 0, \quad \Psi(x, y, z, u) = 0 \quad (4)$$

of orders m, n, p .

Assume the special case in which each of these three functions splits into linear factors. It is then clear that, if the number of common solutions is not infinite, their number must be equal to the product mnp ; for we may set one linear factor of F , one of Φ , and one of Ψ simultaneously equal to zero, and thereby obtain mnp systems of three linear equations, which, if independent of one another, yield the same number of systems of values for the ratios $x : y : z : u$.

At first sight, it is questionable to conclude that in the general case the degree of the final equation of (4) is just as large. But this concern can be entirely removed by the following simple consideration, due to a remark of F. Klein.

Imagine the final equation from equations (4) first to have been formed under the assumption that the coefficients in equations (4) are independent variables. Then the resultant obtained after eliminating x, y is certainly not identically zero, since even for special systems of coefficients, for example when F, Φ, Ψ split into linear factors, it does not vanish identically. We therefore obtain a homogeneous final equation in the two unknowns z, u of some degree, which is to be determined.

If the coefficients are specialized in any way, it is possible that the final equation becomes identically satisfied; but if this case does not occur, then its order cannot change, since all its terms have the same order. Since under the specialization assumed above, expressed by the splitting of the functions F, Φ, Ψ , the identical vanishing does not occur but instead a final equation of order mnp appears, this must also be the degree of the final equation in the general case.

Thus Bezout's theorem is proved for the case of three homogeneous quaternary equations.

That we have restricted ourselves here to three equations was done solely in the interest of simplicity. The extension required for the case of n equations in $n + 1$ homogeneous unknowns follows of itself and may be left to the reader.

For our general problem we shall only add that these considerations show that the problem of algebra, insofar as it concerns the solution of algebraic equations of any kind, is thereby reduced to the treatment of a chain of equations each with one unknown.

§ 51. Decomposable and indecomposable functions

Among integral rational functions of several variables one must distinguish decomposable and indecomposable functions.

An integral rational function of any variables is called decomposable if it can be represented as a product of at least two integral rational functions of the same variables; if it cannot be represented as such a product, it is called indecomposable.

If a function is decomposable, then the degree of each factor is lower than the degree of the function itself. A linear function is therefore always indecomposable, and every integral rational function can be decomposed into a finite number of indecomposable functions.

We say that an integral function W is divisible by another w if a third integral function w' exists such that

$$W = ww'.$$

It follows that if U is divisible by w and V is any integral function, then the product UV is also divisible by w ; and if $U, U', U'', \dots$ are divisible by w , while $V, V', V'', \dots$ are arbitrary integral rational functions, then $UV + U'V' + U''V'' + \dots$ is also divisible by w .

If W is divisible by w , one also says that w divides W .

Two integral rational functions U, V that are not divisible by one and the same integral rational function are called relatively prime, or prime to one another.

We prove the following theorem:

- I. If U, V, v are integral rational functions of any variables, if U and v are relatively prime, and if UV is divisible by v , then V is divisible by v .

This theorem corresponds exactly to a familiar fundamental theorem from the theory of integers: if a product of two integers is divisible by a third integer that is prime to one factor, then the other factor must be divisible by that integer.

We prove it by complete induction. If U, V, v depend on only one variable x , the theorem is correct; for by § 6 (Theorem I), when U and v are relatively prime, one can determine two other integral functions P and p of x such that

$$PU + pv = 1.$$

Multiplication by V gives

$$PUV + pVv = V,$$

and from this it is seen that if UV is divisible by v , then V must also be divisible by v .

We now assume that the theorem to be proved has been established for functions of n or fewer variables, and derive from this its validity for functions of $n + 1$ variables.

For this it is necessary to draw from the theorem I, assumed to be true, some consequences whose final result will give the validity of the theorem for the next higher number of variables.

If v is an indecomposable function, then another function U of the same variables is either relatively prime to v or divisible by v . It follows by I that a product UV of two functions can be divisible by v only when one of the two factors is divisible by v . The same holds for a product of several functions, and thus from I follows the theorem:

- II. A product of several integral rational functions is divisible by an indecomposable function v only when at least one of the factors of the product is divisible by v .

If an integral rational function U is decomposed into indecomposable factors in two ways,

$$U = vv'v'' \dots = ww'w'' \dots,$$

then by II at least one of the factors $v, v', v'', \dots$ must be divisible by w , say v . But since v too is indecomposable, v can differ from w only by a constant factor.

Thus, if c is this constant factor,

$$cv'v'' \dots = w'w'' \dots,$$

from which it follows that one of the functions $v', v'', \dots$, say v' , is divisible by w' and hence differs from w' only by a constant factor.

We therefore obtain as a second consequence of Theorem I:

III. An integral rational function can, apart from constant factors, be decomposed into indecomposable factors in only one way.

From this arises the concept of the greatest common divisor of two or more integral rational functions $U, V, \dots$. By this one understands the product of all indecomposable factors that occur in the decompositions of each of the functions $U, V, \dots$, or the function of highest possible degree that divides all the functions $U, V, \dots$. By III this function is completely determined, up to a constant factor, for every system of functions $U, V, \dots$.

Several functions $U, V, W, \dots$ are called relatively prime if no two of them have a common divisor.

All these definitions and theorems are exactly analogous to well-known theorems of elementary number theory, where they arise as consequences of the algorithm of the greatest common divisor; only here the indecomposable functions take the role of prime numbers.

We record one more such theorem:

IV. If u, v are relatively prime and if Uu, Uv are divisible by w , then U is divisible by w .

For if one decomposes the integral rational functions U, u, v, w into their indecomposable factors, then some factor of w , since it cannot occur in both u and v , must occur in U . Removing it from U and w , one argues in the same way for the next factor of w , and so on.

Now consider integral rational functions of one variable t ,

$$f(t) = u_0t^m + u_1t^{m-1} + \dots + u_{m-1}t + u_m,$$

whose coefficients are integral rational functions of n variables x , independent of t .

If the coefficients $u_0, u_1, \dots, u_m$ have no common divisor, then $f(t)$ is called primitive; otherwise the greatest common divisor of the coefficients $u_0, u_1, \dots, u_m$ is called the divisor of the function $f(t)$ (compare § 2).

Still assuming the validity of I, we conclude:

V. The product of two primitive functions $f(t)$ and $\varphi(t)$ is again a primitive function; and the divisor of a product of two imprimitive functions is equal to the product of the divisors of the two factors.

The proof is exactly as for the corresponding theorem in § 2.

Let

$$\begin{aligned} f(t) &= u_0t^m + u_1t^{m-1} + \dots + u_m, \\ \varphi(t) &= v_0t^\mu + v_1t^{\mu-1} + \dots + v_\mu \end{aligned} \tag{1}$$

be two primitive functions, and let

$$F(t) = U_0t^{m+\mu} + U_1t^{m+\mu-1} + \dots + U_{m+\mu} \tag{2}$$

be their product. Then, if r and s denote any two indices from the rows $0, 1, \dots, m$ and $0, 1, 2, \dots, \mu$ (§ 2),

$$U_{r+s} = u_rv_s + u_{r-1}v_{s+1} + \dots + u_{r+1}v_{s-1} + \dots \tag{3}$$

Now let w be any indecomposable function that does not divide all the u_r and also does not divide all the v_s . We choose r and s in (3) so that u_r is the first u not divisible by w , so that $u_{r-1}, u_{r-2}, \dots$ are divisible by w , and similarly v_s is the first v not divisible by w . Then U_{r+s} cannot be divisible by w , since all terms except the first are divisible by w ; that is, $F(t)$ is primitive.

It follows immediately, if p and q are arbitrary integral rational functions of the x , that pq is the divisor of the product of the two imprimitive functions $pf(t)$ and $q\varphi(t)$, which proves the second part of theorem V. It follows further:

VI. If an integral rational function $F(t)$ of the $n+1$ variables x and t is decomposable into two factors that are integral with respect to t and at least rational with respect to the x , then it is also decomposable into two functions that are integral and rational in x and t .

For by hypothesis there is an integral rational function w of the x alone, and two integral rational functions of x and t , $f_1(t), \varphi_1(t)$, such that

$$wF(t) = f_1(t)\varphi_1(t).$$

By (V), w must divide the product of the divisors of $f_1(t)$ and $\varphi_1(t)$, and hence we also obtain

$$F(t) = f(t)\varphi(t),$$

where $f(t), \varphi(t)$ are again integral rational functions of x and t , and with respect to t have the same degrees as $f_1(t)$ and $\varphi_1(t)$, as was to be proved.

From this we conclude further that two functions $F(t), f(t)$ which, considered as integral rational functions of the $n+1$ variables x and t , are relatively prime in the sense defined above, must also prove to be relatively prime when considered as functions of t alone and treated by the algorithm of the greatest common divisor (§ 6). For if they had a common divisor that was integral with respect to t but fractional with respect to the x , then one could determine an integral function T of x and t and an integral function P of the x alone such that

$$PF(t) = TF_1(t), \quad Pf(t) = Tf_1(t),$$

where F_1, f_1 are integral functions without a common divisor. By IV, P would have to occur in the divisor of T , and $F(t)$ and $f(t)$ could not be relatively prime. Therefore by § 6 one can determine two integral rational functions Q, q of x and t , and an integral rational function X of the x alone, such that the identity

$$QF(t) + qf(t) = X \tag{4}$$

holds.

Now let $\Phi(t)$ be another integral rational function of x and t . From (4), by multiplication with $\Phi(t)$, one obtains

$$Q\Phi(t)F(t) + q\Phi(t)f(t) = X\Phi(t),$$

and if $\Phi(t)F(t)$ is divisible by $f(t)$, then $X\Phi(t)$ is consequently also divisible by $f(t)$.

Thus, if $\varphi(t)$ and $\psi(t)$ again denote two integral functions of x and t , we may put

$$\begin{aligned} X\Phi(t) &= \varphi(t)f(t), \\ F(t)\Phi(t) &= \psi(t)f(t). \end{aligned} \tag{5}$$

Multiplying the second of these equations by X and substituting from the first the expression $\varphi(t)f(t)$ for $X\Phi(t)$, one may cancel $f(t)$ and obtains

$$\varphi(t)F(t) = X\psi(t).$$

By IV, X must therefore occur both in the divisor of $\varphi(t)f(t)$ and in the divisor of $\varphi(t)F(t)$. Since the functions $f(t)$ and $F(t)$, and hence also their divisors, are relatively prime, X must occur in the divisor of $\varphi(t)$. Thus, putting

$$\varphi(t) = X\varphi_1(t),$$

one obtains from (5)

$$\Phi(t) = \varphi_1(t)f(t),$$

that is, $\Phi(t)$ is divisible by $f(t)$. Hence:

VII. If $F(t)$ and $f(t)$ are relatively prime and $\Phi(t)F(t)$ is divisible by $f(t)$, then $\Phi(t)$ is divisible by $f(t)$.

But this is nothing other than Theorem I for functions of $n + 1$ variables, and so I is proved in general.

§ 52. Tschirnhausen transformation

We now make another important application of the theory of symmetric functions and the formation of resultants, namely to the transformation of algebraic equations due to Tschirnhausen.¹

Let

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n \quad (1)$$

be any function of the n th degree, and let

$$y = \frac{\chi(x)}{\psi(x)} \quad (2)$$

be any rational function, integral or fractional, of x , subject only to the assumption that $f(x)$ and $\psi(x)$ are to have no common divisor. By the concluding theorem of § 6 one can then determine functions $F(x)$ and $\varphi(x)$, the latter of degree at most $n - 1$, such that

$$\chi(x) = F(x)f(x) + \varphi(x)\psi(x), \quad (3)$$

and consequently, as soon as x is put equal to a root of the equation

$$f(x) = 0, \quad (4)$$

one has

$$\frac{\chi(x)}{\psi(x)} = \varphi(x).$$

Since we shall consider the values of y only for those values of x that are roots of (4), we lose no generality if from the outset we put

$$y = \varphi(x),$$

therefore equal to an integral function of degree $n - 1$. Thus let

$$y = \varphi(x) = \alpha_0 + \alpha_1x + \alpha_2x^2 + \cdots + \alpha_{n-1}x^{n-1}, \quad (5)$$

where the coefficients α remain for the moment wholly undetermined.

Let the roots of equation (4) be denoted by

$$x_1, x_2, \dots, x_n. \quad (6)$$

Then from (5) there arise the corresponding values of y :

$$y_1 = \varphi(x_1), \quad y_2 = \varphi(x_2), \dots, y_n = \varphi(x_n), \quad (7)$$

and the product

$$\Phi(y) = (y - y_1)(y - y_2) \cdots (y - y_n) \quad (8)$$

¹Acta eruditorum, Leipzig 1683.

is a symmetric function of the quantities (6), and can therefore be expressed rationally through the coefficients of $f(x)$.

Represented in this way, let $\Phi(y)$ have the expression

$$\Phi(y) = y^n + b_1 y^{n-1} + b_2 y^{n-2} + \cdots + b_{n-1} y + b_n. \quad (9)$$

The coefficients b_ν are determined by the y_i , according to § 7, by means of the formulas

$$b_1 = -\sum y_1, \quad b_2 = \sum y_1 y_2, \quad b_3 = -\sum y_1 y_2 y_3, \dots,$$

from which it follows that b_ν is an integral rational and homogeneous function of degree ν in the variables $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$.

$\Phi(y)$ is nothing other than the resultant of the two equations

$$f(x) = 0, \quad \varphi(x) - y = 0, \quad (10)$$

and conversely, by § 49, if y becomes equal to one of the values $y_1, y_2, \dots, y_n$, one can also express the common root of the two equations (10) rationally through the coefficients of (10), that is, rationally through a_i, α_i, y , provided that only one such common root is present.

This will happen when the n values $y_1, y_2, \dots, y_n$ are distinct. Then one obtains an expression

$$x = \beta_0 + \beta_1 y + \beta_2 y^2 + \cdots + \beta_{n-1} y^{n-1}$$

that is valid when one of the values (6) is put for x and the corresponding value (7) for y . The quantities β_i are rationally composed from the a_i and α_i .

If, however, several of the values $y_1, y_2, \dots, y_n$ coincide, then determining the corresponding values of x still requires solving an equation of the corresponding degree. But this degree is always less than n , since the function $\varphi(x)$ can take the same value y for at most $n - 1$ values of x . Thus, when the equation

$$\Phi(y) = 0 \quad (11)$$

has been completely solved, the equation (1) has thereby also been completely solved if (11) has n distinct roots, or at least reduced to the solution of an equation of lower degree if (11) has equal roots.

Accordingly we regard (11) as a transformation of equation (1), and the formation of (11) from (1) is called the Tschirnhausen transformation of equation (1).

The purpose of such a transformation is to determine the coefficients α_i so that $\Phi(y)$ obtains a simpler form than $f(x)$.

The coefficients b_ν can be formed when the power sums of the y_i are known (§ 42).

Let the sums of the m th powers of the x_i be denoted by s_m , and those of the y_i by σ_m , so that

$$s_m = Sx_i^m, \quad \sigma_m = Sy_i^m.$$

Then σ_m can be formed from the s_m in the following way. First expand y^m by the polynomial theorem (§ 12), putting

$$y^m = \sum \frac{\Pi(m)}{\Pi(\nu_0)\Pi(\nu_1)\cdots\Pi(\nu_{n-1})} \alpha_0^{\nu_0} \alpha_1^{\nu_1} \cdots \alpha_{n-1}^{\nu_{n-1}} x^{\nu_1+2\nu_2+\cdots+(n-1)\nu_{n-1}}, \quad (12)$$

where the sum extends over all non-negative integral values of $\nu_0, \nu_1, \dots, \nu_{n-1}$ satisfying the condition

$$\nu_0 + \nu_1 + \cdots + \nu_{n-1} = m. \quad (13)$$

If in (12) one then puts $x_1, x_2, \dots, x_n$ for x and forms the sum of the values thereby obtained for $y_1^m, y_2^m, \dots, y_n^m$, it follows that

$$\sigma_m = \sum \frac{\Pi(m)}{\Pi(\nu_0)\Pi(\nu_1)\cdots\Pi(\nu_{n-1})} s_{\nu_1+2\nu_2+\cdots+(n-1)\nu_{n-1}} \alpha_0^{\nu_0} \alpha_1^{\nu_1} \cdots \alpha_{n-1}^{\nu_{n-1}}. \quad (14)$$

Thus σ_m is represented as an integral homogeneous function of order m in $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$. A somewhat simplified notation is obtained from the second representation of forms (§ 15). Let $\mu_1, \mu_2, \dots, \mu_m$ run independently through the sequence of numbers $0, 1, \dots, n-1$. If in one combination the index 0 occurs ν_0 times, the index 1 occurs ν_1 times, the index 2 occurs ν_2 times, and so on, the index $n-1$ occurs ν_{n-1} times, then

$$\mu_1 + \mu_2 + \dots + \mu_m = \nu_1 + 2\nu_2 + \dots + (n-1)\nu_{n-1},$$

and accordingly the expression for σ_m can be represented as

$$\sigma_m = \sum_{\mu} s_{\mu_1 + \mu_2 + \dots + \mu_m} \alpha_{\mu_1} \alpha_{\mu_2} \dots \alpha_{\mu_m}. \quad (15)$$

For example,

$$\begin{aligned} \sigma_1 &= \alpha_0 s_0 + \alpha_1 s_1 + \dots + \alpha_{n-1} s_{n-1}, \\ \sigma_2 &= \sum_{i,k} s_{i+k} \alpha_i \alpha_k, \\ \sigma_3 &= \sum_{h,i,k} s_{h+i+k} \alpha_h \alpha_i \alpha_k, \end{aligned} \quad (16)$$

where h, i, k run through the sequence of numbers $0, 1, 2, \dots, n-1$.

§ 53. Application to the cubic and biquadratic equations

The aim that Tschirnhausen already had in view in this transformation was to determine the substitution coefficients $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$ so that in the transformed equation $\Phi(y) = 0$ so many terms vanish that the equation becomes solvable. This is easily achieved for equations of the third and fourth degrees, although it does not seem simple, and not possible without extensive calculations, to derive by this method the formulas otherwise known.

First take the cubic equation

$$x^3 + ax^2 + bx + c = 0, \quad (1)$$

which we transform by the substitution

$$y = \alpha + \beta x + \gamma x^2. \quad (2)$$

In order to reduce the equation for y ,

$$y^3 + Ay^2 + By + C = 0, \quad (3)$$

to a pure cubic equation, we have to determine the ratios $\alpha : \beta : \gamma$ from the two equations

$$A = 0, \quad B = 0, \quad (4)$$

of which the first is linear and the second quadratic.

Equations (4) are equivalent to

$$\sigma_1 = 0, \quad \sigma_2 = 0, \quad (5)$$

and hence, by (16) of the preceding paragraph, give

$$\alpha s_0 + \beta s_1 + \gamma s_2 = 0,$$

$$\alpha^2 s_0 + 2\alpha\beta s_1 + 2\alpha\gamma s_2 + \beta^2 s_2 + 2\beta\gamma s_3 + \gamma^2 s_4 = 0,$$

where $s_0 = 3$. If the latter equation is multiplied by s_0 and the square of the former is subtracted, one obtains

$$\beta^2(s_0 s_2 - s_1^2) + 2\beta\gamma(s_0 s_3 - s_1 s_2) + \gamma^2(s_0 s_4 - s_2^2) = 0. \quad (6)$$

The discriminant of this quadratic equation,

$$(s_0s_3 - s_1s_2)^2 - (s_0s_2 - s_1^2)(s_0s_4 - s_2^2),$$

when expanded gives

$$-s_0(s_0s_2s_4 - s_0s_3^2 - s_1^2s_4 - s_2^3 + 2s_1s_2s_3),$$

and hence, if D denotes the discriminant of the cubic equation (1) [§ 46, (9)], is equal to $-3D$. The solution of (6) therefore gives

$$\frac{\beta}{\gamma} = \frac{-(s_0s_3 - s_1s_2) + \sqrt{-3D}}{s_0s_2 - s_1^2}. \quad (7)$$

The equation (3) has thereby been reduced to a pure equation, whose solution requires only the extraction of a cube root. The choice of sign for the square root occurring in (7) is left to us.

To use the Tschirnhausen transformation for the biquadratic equation, one may proceed, for instance, by determining $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ in the transformed equation

$$y^4 + b_1y^3 + b_2y^2 + b_3y + b_4 = 0 \quad (8)$$

so that

$$b_1 = 0, \quad b_3 = 0. \quad (9)$$

Then (8) passes into a quadratic equation for y^2 , from whose roots the square roots must still be drawn; thus after the equations (9) have been solved, a square root has still to be drawn twice.

If by means of the first equation (9) one eliminates α_0 from the second, then a homogeneous cubic equation in $\alpha_1, \alpha_2, \alpha_3$ arises. One may therefore choose one of the two ratios $\alpha_1 : \alpha_2 : \alpha_3$ arbitrarily, for example put $\alpha_3 = 0$, and then obtains a cubic equation to determine the other. Thus the solution of the biquadratic equation has been reduced to that of a cubic and to square roots.

§ 54. The Tschirnhausen transformation of the equation of the fifth degree

If one wants to apply the Tschirnhausen transformation to the equation of the fifth degree

$$x^5 + a_1x^4 + a_2x^3 + a_3x^2 + a_4x + a_5 = 0, \quad (1)$$

one must put

$$y = \alpha_0 + \alpha_1x + \alpha_2x^2 + \alpha_3x^3 + \alpha_4x^4, \quad (2)$$

and obtains the transform

$$y^5 + b_1y^4 + b_2y^3 + b_3y^2 + b_4y + b_5 = 0. \quad (3)$$

The most immediate thought would be to determine the ratios of the five unknowns α so that

$$b_1 = 0, \quad b_2 = 0, \quad b_3 = 0, \quad b_4 = 0 \quad (4)$$

would hold, whereby (3) would be brought back to a pure equation; this was probably also the aim Tschirnhausen had in view. But among the four equations (4), the first is linear, the second quadratic, the third of the third degree, and the fourth of the fourth degree. The final equation derivable from them will therefore be, by § 50, of degree $2 \cdot 3 \cdot 4 = 24$, and hence no benefit for the solution of the equation of the fifth degree can be drawn from it. One therefore seeks first to satisfy only the two equations

$$b_1 = 0, \quad b_2 = 0, \quad (5)$$

whereby equation (3) passes into a form that, following F. Klein's lectures on the icosahedron, is called a principal equation of the fifth degree.

To satisfy the two equations (5), we have at our disposal the four ratios $\alpha_1 : \alpha_2 : \alpha_3 : \alpha_4$, and we can still use these quantities in many ways to simplify the equation of the fifth degree.

We shall return to this subject more fully in a later section, and confine ourselves here to indicating the aims in general. For clarity we use a geometrical formulation, which, however, is not essential for understanding the algebraic theory as it will be given later.

If, with the help of the linear equation $b_1 = 0$, we express one of the five quantities α , say α_0 , in terms of the others, then we may regard the four remaining quantities $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ as homogeneous coordinates of a point in space. The equation $b_2 = 0$ then represents a surface of the second order, $b_3 = 0$ a surface of the third order, $b_4 = 0$ a surface of the fourth order, and $b_5 = 0$ a surface of the fifth order.

If one wanted to set b_2, b_3, b_4 simultaneously equal to zero and thereby reduce the given equation to a pure one, then one would have to determine one of the 24 intersection points of three surfaces of the second, third, and fourth orders; this depends on an equation of the 24th degree.

Instead of proceeding in this way, one seeks to determine a straight line on the surface $b_2 = 0$. On every surface of the second order there is one or two systems of such straight lines, real or imaginary, and one can determine one of these lines by square roots, as will be explained later. The determination of an intersection point of one of these straight lines with the surface $b_3 = 0$ then leads to an equation of the third degree. If, moreover, by determining a common factor of the α one makes the coefficient $b_4 = 1$, the equation for y obtains the form

$$y^5 + y + b = 0, \quad (6)$$

so that y is determined as a function of one variable b .

This equation form is usually named after the English mathematician Jerrard, who made it known in 1834. It had, however, already been published much earlier, in 1786, by E. J. Bring in Lund.² We shall therefore call it the Bring-Jerrard form.

Just as well, one could make $b_4 = 0$ by an equation of the fourth degree and would obtain a second normal form of the equation of the fifth degree:

$$y^5 + y^2 + b = 0, \quad (7)$$

which has the same justification as the Bring-Jerrard form.

For the solution of the equation of the fifth degree, of course, nothing is gained by these considerations. Their use consists in reducing the equation of the fifth degree to a normal form depending on only one variable coefficient. This can be achieved in infinitely many different ways, and is achieved most simply, namely without solving an equation of the third or fourth degree, if one regards the equation of the fifth degree $b_5 = 0$ itself as a transformation of the given equation of the fifth degree. For once the equation $b_5 = 0$ has been solved, one of the five values of y is equal to 0, and the given equation of the fifth degree is at the same time solved.

²Compare F. Klein, Vorlesungen über das Ikosaeder, p. 143.